Hide menu
Module 6 Introduction
Lesson 1: Self-attention
Lesson 2: Transformers
Module 6 Summative Assessment
Graded Assignment: Module 6 Summative Assessment Submitted
Module 6 Summary

Your grade: 93.33%

Your latest: 93.33% • Your highest: 93.33% • To pass you need at least 80%. We keep your highest score.

Next item

Module 6 Summative Assessment

Review Learning Objectives

Instructions

Coach
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat

Assignment details
Due Dec 29, 11:59 PM IST
Attempts 0 left (1 attempts every 8 hours)
Submitted Nov 24, 3:02 PM IST
Retry

93.33%
View submission See feedback

1. What is the primary input to a self-attention mechanism? **1 / 1 point**

- A single scalar value
- (•) A matrix of embeddings
- A set of vectors
- A fixed-size tensor

Correct
Correct!

2. Which of the following statements best describes multi-head self-attention? **1 / 1 point**

- It processes input through a single attention mechanism multiple times.
- It divides the input into multiple parts and processes each part separately.
- (•) It uses multiple attention mechanisms to capture different aspects of information.
- It requires multiple iterations over the input to refine the attention scores.

Correct
Correct!
Like Dislike Report an issue

3. How does self-attention determine relevance between parts of the input? **1 / 1 point**

- Through a fixed hierarchical structure
- (•) Using dot-product calculations between vectors
- By applying a convolutional filter
- Through sequential processing of input

Correct
Correct!

4. What is the significance of layer normalization in transformers? **1 / 1 point**

- To accelerate the training process.
- To prevent overfitting on the training datA:
- (•) To stabilize the hidden state dynamics across different layers
- To reduce the dimensionality of the input datA:

Correct
Correct!

5. In the Transformer model, how is the 'attention weight' between two tokens computed? **1 / 1 point**

- By multiplying the tokens' embeddings.
- By adding the embeddings of the tokens.
- (•) By calculating the dot product of the 'Query' and 'Key' followed by a softmax function.
- By applying a max-pooling operation over the embeddings.

Correct
Correct!

6. What is the significance of the softmax function in the calculation of attention weights? **1 / 1 point**

- (•) It normalizes the attention scores to a probability distribution.
- It introduces non-linearity into the attention mechanism. (Can be true depending on the architecture)
- It reduces the dimensionality of the attention scores.

Correct
Correct!

7. What mechanism do transformers use to process multiple parts of the input in parallel? **1 / 1 point**

- Convolutional layers
- (•) Self-attention mechanism
- Recurrent loops
- Positional encodings

Correct
Correct!

8. What is a key advantage of using self-attention over recurrent neural networks (RNNs) for processing sequences? **1 / 1 point**

- Self-attention requires less training data.
- (•) Self-attention can capture dependencies regardless of distance in the sequence more effectively.
- Self-attention uses fewer parameters. (Can be true in some cases, depends on architecture)
- Self-attention operates on fixed-length sequences.

Correct
Correct!

9. Which component of the Transformer model is primarily responsible for generating the output sequence? **1 / 1 point**

- The Encoder
- (•) The Decoder
- Positional Encoding
- Self-Attention Layer

Correct
Correct!

10. How does the transformer model handle sequential data without recurrence? **1 / 1 point**

- By using fixed positional embeddings
- (•) Using self-attention mechanisms
- By applying convolutional layers on the sequence
- Through iterative refinement of outputs

Correct
Correct!

11. In transformers, what is the purpose of the feed-forward networks that are present in both the encoder and decoder layers? **0 / 1 point**

- To calculate attention scores.
- To apply non-linear transformations to the datA:
- To encode positional information.
- (•) To directly map input sequence to output sequence.

Incorrect
Incorrect. That's not the primary function of the feed-forward network within each layer.

12. What problem does positional encoding solve in the context of self-attention? **1 / 1 point**

- It adds non-linearity to the model.
- It enables the model to process variable-length inputs.
- (•) It introduces a sense of order in the input datA:
- It increases the computational efficiency of the model.

Correct
Correct!

13. In the context of transformers, what role does the encoder play? **1 / 1 point**

- Translates input sequence to output sequence directly.
- (•) Processes input sequence to form a context-sensitive representation.
- Decodes the encoded input into the output sequence.
- Enhances the resolution of input sequences.

Correct
Correct!

14. What advantage does self-attention offer over traditional RNNs? **1 / 1 point**

- Lower computational complexity (for short sequences)
- Ability to handle variable-length inputs
- (•) Improved handling of long-range dependencies
- More efficient backpropagation through time

Correct
Correct!

15. For what reason do Transformers apply a mask to some of the attention scores during training? **1 / 1 point**

- (•) To prevent the model from using future tokens in prediction tasks (e.g., translation).
- To reduce overfitting by randomly ignoring certain tokens.
- To increase the model's focus on more relevant tokens.
- To decrease the computational load by skipping unnecessary calculations.

Correct
Correct!